Hypothesis Testing with E-values – by A. Ramdas and R. Wang

Reading Group 2026

$$\boxed{\text{Chapter 9 - FDR control using compound e-values Part 3/4}}$$

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In the previous episodes

Th 2.9 (Randomized Markov Inequality)

Fix $\mathbb{P} \in \mathcal{M}_1$. Let Y be a non-negative r.v $U \sim \mathcal{U}[0,1]$ be independent of Y. Then, for any $\alpha > 0$,

$$\mathbb{P}\left(Y \geq \frac{U}{\alpha} \right) = \mathbb{E}^{\mathbb{P}}[\min(\alpha Y, 1)]$$
$$\leq \alpha \mathbb{E}^{\mathbb{P}}[Y]$$

In particular if Y is an e-variable for $\mathcal{P}$, then $\mathbb{P}(Y \geq \frac{U}{\alpha}) \leq \alpha \quad \forall \mathbb{P} \in \mathcal{P}$

In the previous episodes

Def 9.8 (The e-BH procedure)
Suppose that $e_1 \dots e_K$ are realised compound e-values for the hypotheses $(\mathcal{P}_1, \dots \mathcal{P}_K)$. For $k \in \mathcal{K}$, let $e_{[k]}$ be the k-th order statistics of $e_1 \dots e_K$, from the largest to the smallest. The e-BH procedure at level $\alpha \in (0,1)$, rejects all hypotheses with the largest k^* e-values, where

$$k^* := \max\left\{ k \in \mathcal{K} \, , \, \frac{k e_{[k]}}{K} \geq \frac{1}{\alpha} \right\}$$

with the convention $\max(\emptyset) = 0$

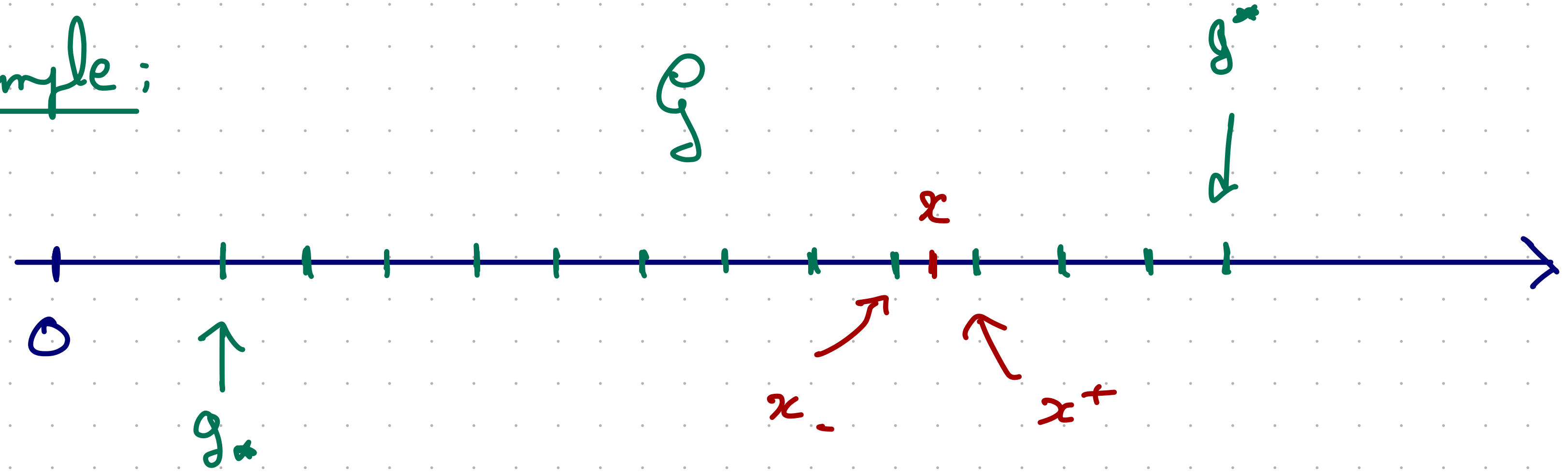
9.5 Stochastic rounding of e-values and randomized e-BH

How to improve e-BH by utilising randomisation.

- $\mathcal{G}$ any closed set $\subseteq [0, \infty]$ (grid)
- $g_* := \inf\{x : x \in \mathcal{G}\}$ $\quad g^* = \sup\{x : x \in \mathcal{G}\}$
- For any $x \in [g_*, g^*]$, let

$$x^+ = \inf\{y \in \mathcal{G} : y \geq x\} \qquad x_- := \sup\{y \in \mathcal{G} : y \leq x\}$$

Exemple :



Def (Stochastic rounding)

Let $x \in [0, \infty]$, its stochastic rounding denoted by $S_{\mathcal{G}}(x)$ is define as follows

$$S_{\mathcal{G}}(x) = x \quad \text{if} \quad x < g_*,\ x > g^*,\ x \in \mathcal{G},\ x = +\infty$$

$$S_{\mathcal{G}}(x) = \begin{cases} x_- & \text{if } U(x) < x \\ x_+ & \text{if } U(x) \geq x \end{cases}$$ otherwise, where $U(x) \sim \mathcal{U}[x_-, x^+]$

Denote by $\mathbb{P}$ the joint distribution of X and $S_{\mathcal{G}}$

Prop 9.24 For any grid $\mathcal{G}$, and any integrable r.v X,

$$\mathbb{E}^{\mathbb{P}}[X] = \mathbb{E}^{\mathbb{P}}[S_{\mathcal{G}}(X)]$$

In particular if X is an e-variable, then $S_{\mathcal{G}}(X)$ is an e-variable

- $S_{\mathcal{G}}(X)$ can be larger than X with (usually) positive proba

The grid can also be random. For instance consider the grid

$$\mathcal{G}(\hat{\alpha}) = \{0, \hat{\alpha}^{-1}, \infty\}$$

We note $S_{\hat{\alpha}}$ for $S_{\mathcal{G}(\hat{\alpha})}$

Prop 9.25 If X is an e-value for $\mathbb{P}$, and $\hat{\alpha}$ taking value in $[0,1]$, possibly depending on X, $S_{\hat{\alpha}}(X)$ is also an e-value, and satisfies $\mathbb{E}^{\mathbb{P}}[S_{\hat{\alpha}}(X)] = \mathbb{E}^{\mathbb{P}}[X]$.

Prop 9.26 If $X \geq \hat{\alpha}^{-1}$ then $S_{\hat{\alpha}}(X) \geq \hat{\alpha}^{-1}$

We want to apply stochastic rounding to e-value, in this context stochastic rounding trade off e-power for power.

Indeed, by Jensen

$$\mathbb{E}^{\mathbb{P}}[\log(S_{\tilde{\alpha}}(E))] \leq \mathbb{E}^{\mathbb{P}}[\log E]$$

So <u>smaller e power</u>

Fix $\alpha \in (0,1)$, and let

$$\alpha_i = \frac{\alpha i}{K} \qquad \mathcal{G}_\alpha := \{\alpha_i^{-1} : i \in \mathcal{K}\} \cup \{0, \infty\}$$

denote the set of possible level that e-BH may reject

If $E \geq \frac{K}{\alpha k}$ for some $k \in \mathcal{K}$ then $S_{\mathcal{G}_\alpha}(E) \geq \frac{K}{\alpha k}$

Thus, an stochastically rounded e-value to $\mathcal{G}_\alpha$ can only improve the power of e-BH

One can view the e-BH procedure as selecting the data dependent threshold

$$\hat{\alpha}^* = \alpha \frac{R^* + 1}{K}$$

and rejecting the i-th hypothesis iff $E_i \geq \frac{1}{\hat{\alpha}^*}$

Now observe that $\frac{1}{\hat{\alpha}^*}$ can only take value on the grid $\mathcal{G}_\alpha$.

Now:

- If we round down each E_i to the grid we get the same discovery as normal e-BH

- If we round up, we do more discoveries!

If we only round up, we lose the e-value property, but by randomising appropriately allow to increase power with positive proba.

Algorithm (Grid-rounded e-BH)

Input: e-values $E_1 \dots E_K$

Rounded: $\left(S_{\mathcal{G}_\alpha}(E_1), \dots S_{\mathcal{G}_\alpha}(E_K) \right)$

Apply the e-BH procedure to $\left(S_{\mathcal{G}_\alpha}(E_1), \dots S_{\mathcal{G}_\alpha}(E_K) \right)$

Th 9.7 For any arbitrarily depend e-values $(E_1 \dots E_K)$ the GeBH procedure ensures FDR $\leq \alpha$ and $\mathcal{D}_{eBH} \subset \mathcal{D}_{GeBH}$. Further, for any $\mathbb{P} \in \mathcal{H}_1$

$$\mathbb{P}\left(\mathcal{D}_{eBH} \subsetneq \mathcal{D}_{GeBH} \right) > 0$$

$$\Leftrightarrow \mathbb{P}\left(\exists k \in [K - k^*] : E_{(k^*+j)} > \frac{K}{\alpha(k^* + k + 1)}, \forall j \in [k] \right) > 0$$

<u>NB</u> This last condition is very weak

Algorithm (Doubly-rounded e-BH)

Input: e-values $E_1 \dots E_K$

Rounded: $\left(S_{\mathcal{G}_\alpha}(E_1), \dots S_{\mathcal{G}_\alpha}(E_K) \right)$

Apply the e-BH procedure to $\left(S_{\mathcal{G}_\alpha}(E_1), \dots S_{\mathcal{G}_\alpha}(E_K) \right)$

Let $\hat{k}^\alpha$ denote the number of rejection above

Define $\hat{\alpha}^* = \alpha \dfrac{\hat{k}^\alpha + 1}{K}$

Reject H_i if $S_{\hat{\alpha}^*}\left(S_{\mathcal{G}_\alpha}(E_i) \right) \geq \dfrac{1}{\hat{\alpha}^*}$

Algorithm (Uniform-randomized e-BH)

Input: e-values $E_1 \dots E_K$, $U \sim \mathcal{U}[0,1]$

Apply the e-BH procedure to

$$\left(\frac{E_1}{U}, \dots, \frac{E_K}{U} \right)$$

Th. 9.29 For any arbitrarily dependent e-values $(E_1, \dots E_K)$ the De-BH and Ue-BH procedure both control FDR at level α. Further:

- $$\mathcal{D}_{eBH} \subseteq \mathcal{D}_{GeBH} \subseteq \mathcal{D}_{DeBH}$$
- $$\mathcal{D}_{eBH} \subseteq \mathcal{D}_{UeBH}$$

Further, $\mathbb{P}(\mathcal{D}_{GeBH} \subsetneq \mathcal{D}_{DeBH}) > 0$

and $\mathbb{P}(\mathcal{D}_{eBH} \subsetneq \mathcal{D}_{UeBH}) > 0$

iff $\mathbb{P}(\exists k \in \mathcal{K}, E_k \in (0, \alpha^{-1})) > 0$

9.6 The closed e-BH procedure

We present a procedure that dominates the e-BH procedure

This section has 4 goals:

Today
- To improve the e-BH procedure, when starting with K e-values, one for each hypothesis
- Recover any given FDR procedure for any problem as an instance of the closed e-BH procedure

Next Week
- To recover and/or improve any given post-hoc FDR procedure using the closed e-BH procedure
- To point out that the techniques developed are applicable to many error metric beyond FDR

9.6.1 Improving the e-BH procedure

- For any $A \subseteq \mathcal{K}$ define $\mathcal{P}_A = \bigcap_{k \in A} \mathcal{P}_k$

The set $\{E_A\}_{A \subseteq \mathcal{K}}$ is called an e-collection if E_A is an e-variable for $\mathcal{P}_A$ for every $A \subseteq \mathcal{K}$.

Example If we have 1 e-value per hypothesis

$$E_A = \frac{1}{|A|} \sum_{k \in A} E_k \qquad A \subseteq \mathcal{K}$$

we call those one the <u>mean e-collection</u>

- For any set $A \subseteq \mathcal{K}$ and $R \subseteq \mathcal{K}$, define

$$\mathrm{FDP}_A(R) = \frac{|A \cap R|}{|R| \vee 1}$$

which is the false discovery proportion of R is the true nulls were A.

Def 9.31 (Closed e-BH procedure)

- Let $\{E_A\}_{A \subseteq \mathcal{K}}$ be an e-collection. Define the candidate discovery

$$\mathcal{C}_\alpha = \left\{ R \subseteq \mathcal{K} : E_A \geq \frac{FDP_A(R)}{\alpha} \ \forall A \subseteq \mathcal{K} \right\}$$

- Define an e-closed procedure as a procedure that output some set $R \in \mathcal{C}_\alpha$.

- Finally, define the closed e-BH procedure ($\overline{\text{e-BH}}$) at level α as the procedure that rejects a largest set in $\mathcal{C}_\alpha$ (any of them)

- Working with the mean e-collection

  Let $R_{[1:k]}$ denotes the indice corresponding to the k largest true e-values with $R_0 = \emptyset$

  The largest set in C_α is $R_{[1:\bar{k}]}$

  where $\bar{k} = \max\{ k \geq 0 : R_{[1:k]} \in C_\alpha \}$

- Lets note $k^\#$ be the number of rejection of the minimally adaptative e-BH procedure (cf last episode)

Th 9.32

Any procedure that reports some $R \in \mathcal{C}_\alpha$ and in particular the closed e-BH procedure, controls the FDR at level α. Moreover, the level post-hoc garantee hold for $\mathbb{P} \subset \mathcal{M}_\alpha$

$$\mathbb{E}^{P}\left[\sup_{\alpha \in (0,1)} \sup_{R \in \mathcal{C}_\alpha} \frac{FDP_{\alpha}(R)}{\alpha} \right] \leq 1$$

Further $\overline{\text{e- BH}}$ always reject a superset of e BH

in fact

$$\bar{k} \geq k^{\#} \geq k^{*}$$

Computat^o time

e-BH $\longrightarrow$ $O(K)$

closed e-BH with mean e-collection $\longrightarrow$ $O(K^3)$

general closed eBH $\longrightarrow$ $O(e^K)$

9.6.2 Recovering any FDR procedure

Th 9.35 Consider an arbitrary FDR controlling procedure $\mathcal{D}$. Then, for each α, there exists an e-collection $\{E_A\}_{A \subseteq \mathcal{K}}$ for which the corresponding closed e-BH procedure recover $\mathcal{D}$ at level α

So every FDR procedure can be written both as

- the e-BH procedure with compound e-values
- the closed e-BH procedure with an e-collection.